Bloch folding, supercells, and what `p2s_map` actually does

Notes for the ACEWorkflow phonon pipeline

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Abstract

Two different operations both get called "folding" in phonon calculations, and conflating them is the usual source of confusion. This note separates them: the *Bloch transform*, which turns an infinite real-space problem into a small $\boldsymbol{q}$-dependent eigenproblem, and *zone folding*, which is what happens to the band structure when you enlarge the unit cell. It then explains where `p2s_map`/`s2p_map` enter (they are index bookkeeping for the first), where the finite supercell introduces an approximation, and why only 16 of the 145 band-path $\boldsymbol{q}$-points in this study are exactly represented.

1 The harmonic crystal and the Bloch transform

Expand the total energy to second order in atomic displacements $\boldsymbol{u}$. Label a primitive cell by its lattice vector $\boldsymbol{R}$, an atom within that cell by i, and a Cartesian direction by α:

$$E = E_0 + \frac{1}{2} \sum_{\boldsymbol{R}\boldsymbol{R}'} \sum_{ij} \sum_{\alpha\beta} u_{i\alpha}(\boldsymbol{R})\, \Phi_{i\alpha,j\beta}(\boldsymbol{R}, \boldsymbol{R}')\, u_{j\beta}(\boldsymbol{R}'), \tag{1}$$

with force constants $\Phi_{i\alpha,j\beta} = \partial^2 E / \partial u_{i\alpha} \partial u_{j\beta}$. The equations of motion are

$$m_i\, \ddot{u}_{i\alpha}(\boldsymbol{R}) = -\sum_{\boldsymbol{R}' j\beta} \Phi_{i\alpha,j\beta}(\boldsymbol{R}, \boldsymbol{R}')\, u_{j\beta}(\boldsymbol{R}'). \tag{2}$$

This is an infinite system of coupled ordinary differential equations — one per atom per direction, over an unbounded crystal.

The fact that rescues it. The crystal is invariant under lattice translations, so Φ cannot depend on *where* the pair sits, only on the separation between the cells:

$$\Phi_{i\alpha,j\beta}(\boldsymbol{R}, \boldsymbol{R}') = \Phi_{i\alpha,j\beta}(\boldsymbol{R}' - \boldsymbol{R}). \tag{3}$$

Equation (2) is then a *convolution* on the lattice, and convolutions diagonalise under Fourier transform. Substituting the plane-wave ansatz

$$u_{j\beta}(\boldsymbol{R}, t) = \frac{1}{\sqrt{m_j}}\, e_{j\beta}(\boldsymbol{q})\, e^{i(\boldsymbol{q}\cdot\boldsymbol{R} - \omega t)} \tag{4}$$

makes every explicit $\boldsymbol{R}$ cancel, leaving a finite eigenproblem at each $\boldsymbol{q}$ separately:

$$\omega^2(\boldsymbol{q})\, e_{i\alpha}(\boldsymbol{q}) = \sum_{j\beta} D_{i\alpha,j\beta}(\boldsymbol{q})\, e_{j\beta}(\boldsymbol{q}), \tag{5}$$

$$D_{i\alpha,j\beta}(\boldsymbol{q}) = \frac{1}{\sqrt{m_i m_j}} \sum_{\boldsymbol{R}} \Phi_{i\alpha,j\beta}(\boldsymbol{R})\, e^{i\boldsymbol{q}\cdot\boldsymbol{R}}. \tag{6}$$

The lattice sum in Eq. (6) is the folding. An infinite problem has collapsed to a $3N_p \times 3N_p$ Hermitian matrix at each $\boldsymbol{q}$, where N_p is the number of atoms in the *primitive* cell. For FCC aluminium $N_p = 1$, so $\boldsymbol{D}(\boldsymbol{q})$ is 3×3 and there are exactly three branches at every $\boldsymbol{q}$: two transverse acoustic and one longitudinal.

In the code this is `src/Phonons/phonon_bands.jl:121--154`. The lattice vector $\boldsymbol{R}$ is built at line 130, the phase $e^{i\boldsymbol{q}\cdot\boldsymbol{R}}$ at lines 139–140, and the mass normalisation $1/\sqrt{m_i m_j}$ at line 145.

2 Why it is called "folding"

The wavevector $\boldsymbol{q}$ is not unique. A reciprocal lattice vector $\boldsymbol{G}$ is defined by $e^{i\boldsymbol{G}\cdot\boldsymbol{R}} = 1$ for every lattice vector $\boldsymbol{R}$, so

$$e^{i(\boldsymbol{q}+\boldsymbol{G})\cdot\boldsymbol{R}} = e^{i\boldsymbol{q}\cdot\boldsymbol{R}} \qquad \Longrightarrow \qquad \boldsymbol{D}(\boldsymbol{q}+\boldsymbol{G}) = \boldsymbol{D}(\boldsymbol{q}). \tag{7}$$

Adding $\boldsymbol{G}$ produces a physically *identical* displacement pattern. So $\boldsymbol{q}$ is meaningful only modulo $\boldsymbol{G}$, and every wavevector outside the first Brillouin zone *folds back* into it. Physically: there is no phonon with a wavelength shorter than the interatomic spacing. Attempting to specify one merely re-describes a longer-wavelength mode, because the atoms only sample the wave at discrete points.

What $\boldsymbol{q}$ means. A wavevector is a *phase relationship between neighbouring cells*, nothing more. At $\boldsymbol{q} = \boldsymbol{0}$ (Γ) every cell moves identically, in lockstep. At a zone boundary, adjacent cells move exactly out of phase. In between, there is a progressive phase shift — a travelling wave of wavelength $2\pi/|\boldsymbol{q}|$.

3 Zone folding: what happens when you enlarge the cell

Now take a supercell of N_{cell}^3 primitive cells instead of the primitive cell itself. Its real-space lattice is coarser by N_{cell} in each direction, so its *reciprocal* lattice is *denser* by the same factor and its Brillouin zone is smaller in volume by N_{cell}^3.

Wavevectors that were ordinary interior points of the primitive zone are now reciprocal lattice vectors *of the supercell*, so by Eq. (7) they fold onto Γ. The three primitive branches, sampled at the N_{cell}^3 commensurate wavevectors, reappear as $3N_{\text{cell}}^3$ branches crammed into the smaller zone (Fig. 1).

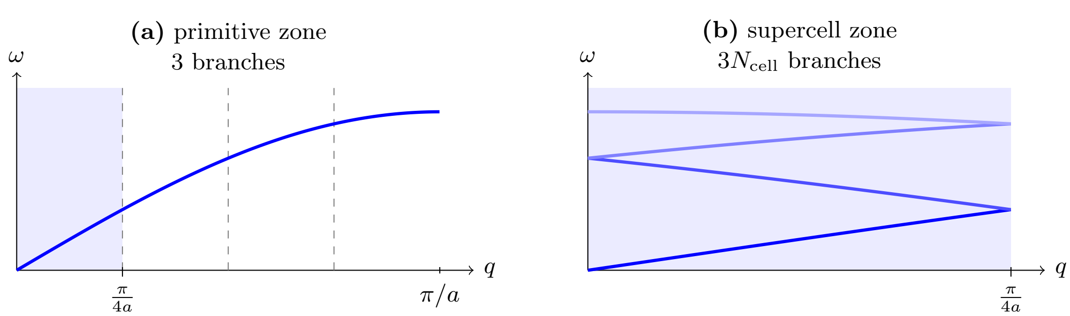

Figure 1: Zone folding in one dimension for $N_{\text{cell}} = 4$. (a) The acoustic branch across the primitive Brillouin zone; dashed lines mark the supercell zone boundaries. (b) The same physical content re-expressed in the $4\times$ smaller supercell zone: each segment of (a) folds back into the shaded region, producing four branches where there was one. No physics changes — only the labelling.

For the cells used in this study: the conventional cubic FCC cell holds 4 atoms, so a $4 \times 4 \times 4$ conventional supercell holds $4 \times 4^3 = 256$ atoms, i.e. 256 primitive cells. Its Γ point therefore contains, superimposed, the primitive modes at all 256 commensurate wavevectors:

$$3 \times 256 = 768 = 3N_{\mathrm{at}}, \tag{8}$$

which is exactly the dimension of the raw supercell Hessian.

The two operations are near-inverses. Table 1 states them side by side. They carry the same physical content whenever $\boldsymbol{q}$ is commensurate; they differ only in bookkeeping.

Table 1: The two senses of "folding", and where each is implemented.

	Bloch transform	Zone folding
direction	real space $\to \boldsymbol{D}(\boldsymbol{q})$	large BZ $\to$ small BZ
cell used	true primitive cell	supercell as its own cell
result (FCC Al, 4^3)	3 branches, $\boldsymbol{q}$ continuous	768 branches at fixed $\boldsymbol{q}$
implementation	`phonon_bands.jl:121`	`get_hessian_phonopy.jl:261`

4 What `p2s_map` and `s2p_map` are

They are the index arithmetic that lets the lattice sum of Eq. (6) be run as a sum over *supercell atoms*. Every supercell atom is a lattice translate of exactly one primitive atom, which partitions the N_s supercell atoms into N_p equivalence classes:

$$\texttt{p2s\_map}[i] : \text{primitive atom } i \longmapsto \text{index of its representative supercell atom,} \tag{9}$$

$$\texttt{s2p\_map}[k] : \text{supercell atom } k \longmapsto \text{index of } \textit{that atom's} \text{ representative, also in supercell space.} \tag{10}$$

Note that *both* take values in supercell index space. `s2p_map` does not return a number in $0, \ldots, N_p - 1$. That is precisely why the two can be compared directly to select the images of a given primitive atom, as at `phonon_bands.jl:128`:

```
s2p_map[k + 1] == j || continue
```

which reads "skip supercell atom k unless it is a periodic image of primitive atom j". The sum that is actually evaluated is

$$D_{i\alpha,j\beta}(\boldsymbol{q}) = \frac{1}{\sqrt{m_i m_j}} \sum_{k\,:\,\texttt{s2p}(k)=j} \boldsymbol{H}\big[\,3\,\texttt{p2s}(i) + \alpha,\ 3k + \beta\,\big]\, e^{i\boldsymbol{q}\cdot\boldsymbol{R}_k}, \qquad \boldsymbol{R}_k = \boldsymbol{r}_k - \boldsymbol{r}_j. \tag{11}$$

The row index is pinned to a representative; the column sweeps over all images. These maps are *not* periodic boundary conditions — they encode translational symmetry, which an infinite crystal possesses regardless of any boundary. In this repository they are constructed by explicit matching at `phonon_bands.jl:81--106`.

5 Where the finite supercell approximates

Equation (6) requires $\Phi(\boldsymbol{R})$ for the infinite crystal, but a supercell calculation with periodic boundaries returns the *image sum*

$$\boldsymbol{H}[i,\, j+\boldsymbol{R}] \;=\; \sum_{\boldsymbol{n}} \Phi\big(\boldsymbol{R} + \boldsymbol{n}\cdot \boldsymbol{L}_{\text{super}}\big), \tag{12}$$

lumping together every periodic copy. The individual $\Phi(\boldsymbol{R})$ is recovered only if Φ has decayed to zero before the supercell boundary, so that exactly one term in Eq. (12) survives. This is the documented precondition at `phonon_bands.jl:52--54` and is the real constraint on N_{cell}.

How large is large enough. For a site potential $E = \sum_i \varepsilon_i$ with cutoff r_c, the Hessian has range $2r_c$, not r_c: the cross term $\partial^2 E/\partial \boldsymbol{r}_a \partial \boldsymbol{r}_b$ is non-zero whenever some site i has *both* a and b as neighbours. With $r_c = 6$ Å the force constants therefore extend to ≈ 12 Å, while a $4\times4\times4$ cell at $a = 4.045$ Å has an edge of 16.2 Å and a half-width of 8.1 Å. Strict image isolation would want an edge exceeding $2 \times 2r_c = 24$ Å, i.e. $N_{\text{cell}} \geq 6$.

This gap is the standard situation in supercell phonon work and the residual terms are far-tail and small, but it is an empirical question, not a provable one. The check that settles it is convergence against a larger cell.

6 Consequence: interpolation and commensurability

Because the force constants are known for a finite set of $\boldsymbol{R}$, evaluating Eq. (6) at an arbitrary $\boldsymbol{q}$ sums that finite set against an arbitrary phase. This is **Fourier interpolation**: exact where $\boldsymbol{q}$ is commensurate with the supercell, since there the finite $\boldsymbol{R}$-set is complete, and interpolated everywhere else, with quality set by how fast Φ decays.

A finite box with periodic boundaries can only host waves that close on themselves — an integer number of wavelengths across the box:

$$\boldsymbol{q}\cdot \boldsymbol{L}_{\text{super}} \in 2\pi\mathbb{Z}. \tag{13}$$

Anything else would be discontinuous where the box wraps onto itself. Converting a primitive-cell fractional wavevector $\boldsymbol{f}$ to conventional-cell fractional coordinates,

$$\boldsymbol{g} = (-f_1 + f_2 + f_3,\;\; f_1 - f_2 + f_3,\;\; f_1 + f_2 - f_3), \tag{14}$$

Eq. (13) becomes the requirement that $N_{\text{cell}}\,\boldsymbol{g} \in \mathbb{Z}^3$.

Applying this to the band path $\Gamma \to X \to U \to L \to \Gamma \to K$ with $[20, 20, 20, 20, 60]$ points per segment: of the 145 sampled wavevectors, only **16 (**11.0%**)** are commensurate with a $4\times4\times4$ cell. Per segment the counts are 5/21, 2/21, 2/21, 3/21 and 4/61. The remaining 129 are interpolated — legitimate, and as good as the decay of Φ, but not independent data.

Why this matters for the constraint. The same restriction applies to any molecular-dynamics cell of the same size. A $4\times4\times4$ MD supercell cannot host, and therefore cannot falsify, the 89% of constrained modes that are incommensurate with it. This is not a defect in the constraint: it means the harmonic constraint acts on a finer $\boldsymbol{q}$-grid than the MD can resolve, so the constraint is doing strictly more than the dynamics can check. A naive committee member carrying imaginary modes at incommensurate $\boldsymbol{q}$ may nonetheless run as a clean FCC crystal in that cell.

Summary

1. Translational symmetry, Eq. (3), makes the equations of motion a convolution; Fourier transforming it gives the dynamical matrix Eq. (6). That sum is the Bloch folding, and it reduces an infinite problem to $3N_p \times 3N_p$ at each $\boldsymbol{q}$.

2. $\boldsymbol{q}$ is defined only modulo a reciprocal lattice vector, so wavevectors fold into the first Brillouin zone. Enlarging the cell shrinks that zone and folds the branches onto each other — the same physics, more branches.

3. `p2s_map`/`s2p_map` are the index bookkeeping for the sum in Eq. (11). They encode translational symmetry, not periodic boundary conditions.

4. The finite supercell enters through Eq. (12): individual force constants are recovered only once Φ has decayed within the box, which needs verifying against a larger cell rather than assuming.

5. Off the commensurate grid, $\boldsymbol{D}(\boldsymbol{q})$ is an interpolation. Only 16/145 band-path points are exact at $N_{\text{cell}} = 4$, which is also why MD in a cell that size cannot test most of the constraint.